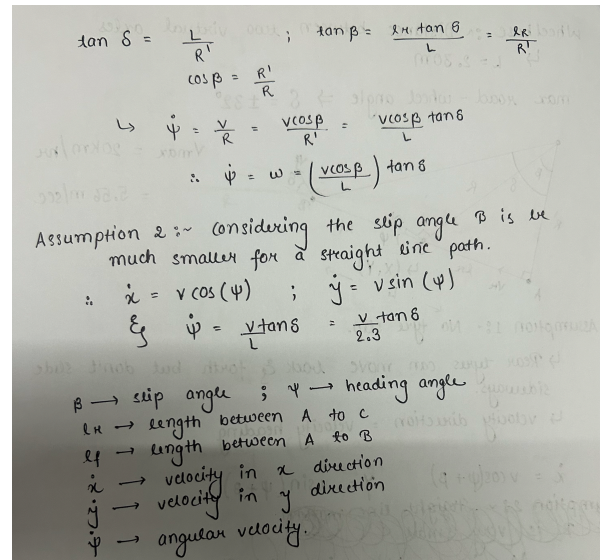
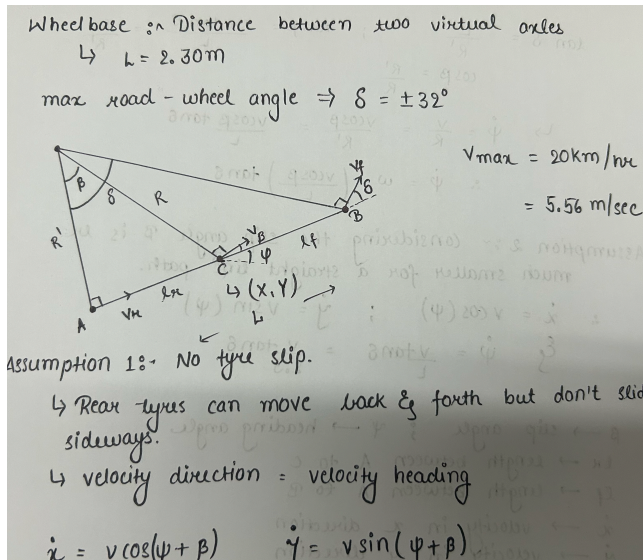


Controls

Q1.)

(A) Planar Vehicle Modeling and Dynamic Model Transition :

(a) This is basically focused on kinematic bicycle model for a car-like vehicle. I am attaching a photo of my calculations as showing calculations with this typed format is very much difficult....



Assumption 1 — Rigid vehicle : The buggy is treated as a rigid body.

Assumption 2 — Flat road : The road is assumed to be approximately flat.

Assumption 3 — No lateral tyre slip : The tyres are assumed to roll without significant sideways sliding.

Assumption 4 — Bicycle approximation : The four wheels are replaced by two equivalent wheels : front-left + front-right $\rightarrow$ one front wheel
 rear-left + rear-right $\rightarrow$ one rear wheel

Assumption 5 — One equivalent steering angle : The two front wheels are represented by a single equivalent steering angle δ .

In reality, the inner and outer front wheels have slightly different steering angles because of Ackermann geometry.

For this model, we use one equivalent road-wheel angle.

Assumption 6 — Rear axle is the reference point : The position : (x, y) represents the centre of the rear axle, not the CG.

Assumption 7 — No significant aerodynamic effects.

Why is this the standard starting model for a low-speed vehicle?

- A simple one line answer would be : "A vehicle is not a point that could move in any direction."
- For example : Consider a point object that can go back & forth but it can move in any direction theoretically. But in the case of buggy, if the buggy is facing north, it cannot move in east and west directions. It has to steer and follow a curved path....

(b) In Part A(a), we assumed that the tyres do not slip sideways. But a real tyre develops a lateral force when the vehicle turns.

For example, when the buggy turns left : The direction in which the tyre is pointing and the direction in which the tyre is actually moving are not exactly the same. That difference is called the slip angle....

Motion Planning

Q1.)

(A) Fundamentals, Applied :

(a) Configuration Space of the Buggy : The configuration space (C-space) is the set of all possible configurations that completely describe the position and orientation of the vehicle.....

For the buggy, we can represent a configuration as :: $q = (x, y, \theta)$
where

- x = position of the buggy along the x - axis
- y = position of the buggy along the y - axis
- θ = heading/orientation of the buggy with respect to the x - axis

Therefore, the buggy has a 3-dimensional configuration space : $C = \mathbb{R}^2 \times S^1$

The x and y dimensions describe where the buggy is located, while θ describes which direction it is facing....

The steering angle δ is not treated as an additional configuration dimension in this simplified planning model. Instead, it is a control variable that determines how the buggy changes its heading. For the buggy,

$$|\delta| \leq 32^\circ$$

and its motion is constrained by the kinematic bicycle model :

$$\dot{x}^* = v \cos\theta \quad \dot{y}^* = v \sin\theta \quad \dot{\theta}^* = (v/L) \tan\delta$$

where $L = 2.30$ m is the wheelbase. This means the buggy cannot arbitrarily change x, y, θ independently.

The buggy is therefore a non-holonomic vehicle: it cannot move sideways instantaneously, and its heading can only change by following a curved trajectory.....

C-space of a point robot : For a point robot moving on a 2D plane, its configuration only needs its position : $q = (x, y)$

Therefore, $C_{\text{point}} = \mathbb{R}^2$ and the point robot has only 2 configuration dimensions.

The point robot does not have a physical orientation that affects its motion. Thus, an ordinary 2D grid such as (x, y) is sufficient to represent its possible configurations....

A normal 2D A* planner considers only the position of the vehicle. Therefore, it can produce a path such as :

forward $\rightarrow$ sharp turn $\rightarrow$ forward

without considering whether the buggy can physically make that turn.

A real buggy cannot turn on the spot. Its heading changes according to

$$\dot{\theta}^* = (v/L) \tan\delta$$

and the steering angle is limited to $|\delta| \leq 32^\circ$

Therefore, the buggy has a minimum turning radius and cannot follow arbitrary paths, even if those paths are collision-free in the 2D map. Car-like motion planning must account for both orientation and the non-holonomic constraints.....

(b) For a car-like vehicle, the minimum turning radius is given by

$$R_{\min} = \frac{L}{\tan(\delta_{\max})}$$

Substituting the values, we get

$$R_{\min} = 3.68\text{m}$$

$$K_{\max} = 1/R_{\min} = 0.272\text{m}^{-1}$$

Example of a path produced by 8-connected A*

Consider a grid with 1 m × 1 m cells. Suppose obstacles force a path containing the following sequence of points:

$$(2, 2) \rightarrow (2, 3) \rightarrow (2, 4) \rightarrow (3, 4) \rightarrow (4, 4)$$

The path initially moves straight upward and then makes a 90° turn to the right.....

A normal 8-connected A* planner considers this path valid because every consecutive grid point is connected and there is no collision with an obstacle.....

However, the path contains an abrupt change in heading : 90° → 0° at the point (2, 4).

This is a sharp corner. The direction changes instantaneously, so the required curvature at the corner is effectively infinite : $K_{A^*} \rightarrow \text{infinity}$

This is much greater than the buggy's maximum possible curvature : $K_{A^*} > K_{\max} = 0.272\text{m}^{-1}$

Therefore, the buggy cannot physically execute this path.....

For example, to make a 90° turn at this minimum radius, the buggy would need to follow a quarter-circle. The required arc length would be : $s = R\theta$ where $\theta = (\pi/2)$ rad.....

Therefore :: $s = 5.78\text{m}$

Hence, instead of making the instantaneous 90° corner proposed by grid A*, the buggy needs approximately 5.78 m of curved path to perform the same 90° change in heading at its minimum turning radius....

A 1 m grid path that turns immediately at a single grid cell therefore does not provide enough distance or curvature for the buggy to perform the manoeuvre....

(c) Hybrid A* is an improved version of the normal grid-based A* algorithm designed especially for vehicles that cannot move freely in every direction....

In normal grid-based A*, the environment is divided into square cells, and the vehicle is usually allowed to move from one cell to another in directions such as up, down, left, and right. The algorithm mainly considers the position (x, y) of the vehicle. However, a car-like vehicle cannot instantly move sideways or change its direction by 90°. It has a limited turning radius and its orientation also matters....

Hybrid A* solves this problem by considering the vehicle's position and orientation :

$$(x, y, \theta)$$

where :

- (x, y) = position of the vehicle
- (θ) = heading/orientation of the vehicle

Instead of simply moving from one grid cell to another, Hybrid A^* generates possible vehicle motions using a car-like motion model. For example, it can consider different steering angles and forward/reverse movements. It then searches for the lowest-cost path while checking whether the vehicle collides with obstacles....

Section - A

Q1.) My short-term goal is to build a strong foundation in mechanical engineering, especially in vehicle dynamics, mechanisms, design and practical problem-solving. I want to gain hands-on experience by working on real engineering problems rather than only studying them theoretically. Through SeDriCa, I also want to understand how mechanical systems interact with sensing, control and autonomous decision-making.....

My long term goal is to apply this knowledge in the challenging engineering problems that come across my future. Eventually I believe that SeDriCa could help me take my first serious step towards this goal.....

Q2.) In my childhood, I usually used to disassemble whole car toys and try to find out the working of that toy...😓 But I would not able to assemble it completely.... I think I was motivated by that memory of my childhood. Further I have seen that how team works and was well influenced by that. I also want to contribute to the working of the team and want to learn as much as possible from seniors....

Q3.) Situation: Main Gate during a busy period

A difficult situation for the autonomous buggy would be near the IIT Bombay Main Gate, especially when there is a mixture of pedestrians, two-wheelers, cars and other vehicles entering or leaving the campus. A pedestrian may suddenly start crossing the road while a two-wheeler is approaching from behind or another vehicle is trying to overtake the buggy. The situation becomes even more difficult when several road users are moving in different directions at the same time.

- (a) Pedestrians may change direction or enter the road suddenly, while two-wheelers and cars may overtake or change lanes. Since the buggy operates in traffic on IIT Bombay's roads, it cannot assume that every road user will follow a predictable path.
- (b) I think this situation would mainly challenge sensing, prediction and planning. The sensors need to correctly detect pedestrians and vehicles and estimate their positions and motion. The prediction system then has to estimate whether a pedestrian or vehicle is likely to enter the buggy's path. Based on this, the planning system must decide whether to continue, slow down or stop. Since the buggy carries passengers, the controller should also avoid sudden braking or steering whenever possible.
- (c) One practical approach would be to create a conservative safety zone around the buggy and reduce its speed whenever the system detects uncertain or potentially conflicting movements. If a pedestrian or vehicle enters this zone, the planner could command a controlled slowdown or stop. A safety driver and mechanical override are also available on the planned vehicle as an additional safety layer.

—
I would like to say one thing that I was not able to complete this assignment due to my work overload in this weekend....

I have participated in IDEAS program and I in that our team has to show customer discovery for the project.... So we were not in campus for almost 3 days....

And as we also represented our IDEAS program idea in the SiH, we had to create ppt and video due to which, I had a very little time....

I had learned many other things but I am also not able to complete the pdf and coding part....
I had also learned the buggy question part 2.... But I was not able to write it on book due to lack of time....
I also had other commitments like AeRoVe tech team, E-cell, TechFest....
But if I got elected, I will leave all these commitment.....